Vijay Barman

Pune, India · vbarman116@gmail.com · linkedin.com/in/barmanjii88 · github.com/Barmanjii

Backend engineer for robot fleets: C++ on the robot, Python and AWS in the cloud, and the operator UI in React and Flutter when needed. Ramp up fast on new or legacy code with AI coding agents, gated by design review and tests.

EXPERIENCE

Peppermint Robotics, Pune · autonomous cleaning robots — Sep 2022 to present

SDE 2, Backend (Senior Design Engineer 2) — Apr 2025 to present

- Migrated production customers from the AWS India region to a US region, with global routing on Lambda and DynamoDB, all in Terraform.
- Redesigned fleet reporting pipelines and storage for faster, more accurate robot reports.
- Built a ROS 2 C++ node that streams robot telemetry and reports to the cloud.
- Shipped the fleet manager to a factory with no internet, as an offline Docker stack with MQTT over mutual TLS.
- Built the fleet manager web app and robot sensor profile editor (React, TypeScript), and mission scheduling in our Flutter app.
- Helped write the team's AI coding agent rules; interview and mentor SDE 1s.

SDE 1, Backend (Software Development Engineer 1) — Dec 2022 to Mar 2025

- Built the fleet dashboard backend (FastAPI, SQLAlchemy, PostgreSQL) with role based access control on OPA.
- Built the backend of our in house update service that deploys software to robots.
- Built Flutter features for the material handling app, including robot discovery on the local network.

Software Engineer Intern — Sep 2022 to Nov 2022

Earlier: Data Analyst (SQL), 2020 to 2021.

SKILLS

- **Languages:** Python, SQL, C++, TypeScript, Dart
- **Backend and data:** FastAPI, PostgreSQL, TimescaleDB, Redis, MQTT
- **Cloud:** AWS (ECS, RDS, S3, Lambda, DynamoDB), Terraform, Docker, Linux
- **Frontend:** React, Flutter
- **Robots:** ROS 2
- **AI:** Claude Code, Cursor

EDUCATION

MCA, Dr. A.P.J. Abdul Kalam Technical University — 2021 to 2023

BCA, Chaudhary Charan Singh University — 2016 to 2019